Extended Data Fig. 8 | Lineage-tracing reference: scLT-seq. a, CCE views of Day 2.5 (1,756 cells) and Day 5 (675 and 755 cells in two replicates). **b,** Clone-transition examples for clone IDs 4, 6, 7, 23 and 32. **c,** Clone-based reference streams and all 19 method configurations for 1,756 cells, colored by cell type. **d,** Per-cell cosine similarity to the clone-based reference. Numbers below method names in the stream panels are the median per-cell reference-direction cosines, shown to three decimal places. Boxes show the median and interquartile range, with 1.5-IQR whiskers and outliers; methods are ordered by median. All 19 configurations use barcode-aligned cells and the same reference coordinates. Cosines were calculated from the displayed per-cell directions. TopicVelo has 1,724 valid cosines; its 32 zero-direction cells remain undefined and are excluded from its boxplot. All other configurations have 1,756 valid cosines.